

Data Structures and Algorithms

No GenAI allowed.

(Use Git version control to track commits, upload java files and git_history.txt file)

Lab 3

- Create a *displayStack* method in the **StackReferenceBased** class that displays or prints out the stack vertically and call this method from the main method in the test class. Indicate which element is at the top. (Very similar to *displayList* from last week).
- Create a method in the test class, public static boolean *isBalanced*(String s), that accepts a string as input and uses a Stack to check if the braces "{}" in the String are balanced.
- Create a menu driven program in the test class that displays the following menu to the user and reads their selection using the Scanner class.

Welcome to StackTest! Please select a number from the list.

1. Push a string on to the stack
2. Pop a string from the stack
3. Peek at the top of the stack
4. Empty the stack
5. Check if a string has balanced brackets.
6. Quit the program

After each selection the stack should be updated and displayed so the user can see the contents. If the user selects the balanced brackets option your program should allow them to enter a string (Scanner class again) and use a stack to determine if the brackets are balanced.